Huawei Access Node IND-ACC-58 Indiranagar Exchange Hardware/Environmental Fault Alarm Handling SOP

Node: IND-ACC-58 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_1008

Description

The Huawei Access Node IND-ACC-58 Indiranagar Exchange is prone to hardware/environmental faults due to the inherent susceptibility of the ASIC memory parity error and line card crash vulnerabilities in the Huawei Access Node architecture. This is primarily attributed to the complex interplay of the Line Card's (LC) role in managing the physical interface and the ASIC's (Application-Specific Integrated Circuit) responsibility for handling the data plane traffic. The power supply redundancy loss, in this case, exacerbates the situation by rendering the node vulnerable to complete system failure. Furthermore, the specific CLI dialect used by Huawei adds an additional layer of complexity, as it necessitates a deep understanding of the vendor-specific syntax and semantics to effectively diagnose and resolve the issue at hand.

Pre-requisite Checks

- Ensure the Indiranagar Exchange site is accessible for remote hands and that the necessary personnel are dispatched to the location.

- Verify the status of the redundant link and ensure it is operational before proceeding.

- Check the power supply redundancy status and ensure it is functioning correctly.

Potential Root Causes

Cause	Probability
ASIC Memory Parity Error	High
Line Card Crash	Medium
Power Supply Redundancy Loss	Low

Diagnostic Commands & Expected Outputs

```
Command: display this system, Expected Output: System information including software version and hardware configuration, Description: This command provides an overview of the system, including the software version and hardware configuration, which can aid in identifying potential issues.

Command: display board info, Expected Output: Board information including firmware version and status, Description: This command provides detailed information about the boards in the system, including the firmware version and status, which can help identify
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potential issues with the line cards or ASICs.

Command: display power supply info, Expected Output: Power supply information including status and redundancy status, Description: This command provides information about the power supplies, including their status and redundancy status, which can aid in identifying potential issues with power supply redundancy.
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Resolution Steps

Step 1: Step: Replace the Faulty ASIC, Command: replace asic 0/0/0, Description: This command replaces the faulty ASIC with a new one, which should resolve the issue if the problem is indeed with the ASIC.

Step 2: Step: Reboot the System, Command: system reboot, Description: This command reboots the system, which can sometimes resolve issues related to software or firmware bugs.

Step 3: Step: Replace the Faulty Line Card, Command: replace line card 0/0/0, Description: This command replaces the faulty line card with a new one, which should resolve the issue if the problem is indeed with the line card.

Post-Check Verification

```
Command: display this system, Expected Output: System information including software version and hardware configuration, Description: This command provides an overview of the system, including the software version and hardware configuration, which can aid in verifying that the issue has been resolved.

Command: display board info, Expected Output: Board information including firmware version and status, Description: This command provides detailed information about the boards in the system, including the firmware version and status, which can help verify that the issue has been resolved.

Command: display power supply info, Expected Output: Power supply information including status and redundancy status, Description: This command provides information about the power supplies, including their status and redundancy status, which can aid in verifying that the issue has been resolved.
```

Roll-back Procedures

- Step: Revert the ASIC Replacement, Command: undo replace asic 0/0/0, Description: This command reverts the ASIC replacement, which can be useful if the replacement ASIC is not functioning correctly.

- Step: Revert the System Reboot, Command: undo system reboot, Description: This command reverts the system reboot, which can be useful if the reboot did not resolve the issue.

- Step: Revert the Line Card Replacement, Command: undo replace line card 0/0/0, Description: This command reverts the line card replacement, which can be useful if the replacement line card is not functioning correctly.